PHYSICS-KM-01

Topics Covered: *Distance, Displacement, speed, velocity and Acceleration*

Total Count of Questions: *22*

Expected Time of Attempt by Students: *110 min*

1. A man has to go 50 m due north, 40 m due east and 20 m due south to reach a field. (a) What distance he has to walk to reach the field? (b) what is his displacement from his house to the field?

2. A particle starts from the origin, goes along the X-axis to the point (20 m, 0) and then returns along the same line to the point (–20m, 0). Find the distance and displacement of the particle during the trip.

3. It is 260 km from a Patna to Ranchi by air and 320 km by road. An aeroplane takes 30 minutes to go from Patna to Ranchi whereas a deluxe bus takes 8 hours.
(a) Find the average speed of the plane.
(b) Find the average speed of the bus.
(c) Find the average velocity of the plane.
(d) Find the average velocity of the bus.

4. When a person leaves his home for sightseeing by his car, the meter reads 12352 km. When he returns home after two hours the reading is 12416 km. (a) What is the average speed of the car during this period? (b) What is the average velocity?

5. An athlete takes 2.0 s to reach his maximum speed of 18.0 km/h. what is the magnitude of his average acceleration?

6. A man walks at a speed of 6 km/hr for 1 km and 8 km/hr for the next 1 km. What is the average speed for the walk of 2 km?

7. The I.Sc. lecture theater of a college is 40 ft wide and has a door at a corner. A teacher enters at 12.00 noon through the door and makes 10 round along the 40 ft wall back and forth during the period and finally leaves the class-room at 12:50 p.m. through the same door. Compute his average speed and average velocity.

8. The velocity of a car changes from 72 km/hr due east to 72 km/hr due north in 10 seconds. What is the average acceleration of the car over this duration of time?

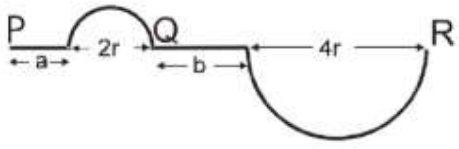
9. A person walks with constant speed of 5 km/hr. He walks for 1 hr. due east, then for 2 hrs. due north, then for 1 hr. again due east and finally for $\sqrt{2}$ hr. due south west.
(a) What is the displacement of the person?
(b) What is the total distance travelled by him?

10. A farmer has to go 500 m due north, 400 m due east and 200 m due south to reach his field. If he takes 20 minutes to reach the field,
(a) What distance he has to walk to reach the field?
(b) What is his displacement from his house to the field?
(c) What is the average speed of farmer during the walk?
(d) What is the average velocity of farmer during the walk?

11. A cyclists moves with constant speed 5 m/s along eastward for 2 seconds. Then along southward for 2 seconds, then he moves along west for one second and finally along North-west for $\sqrt{2}$ seconds. Find
(a) Distance and displacement of cyclist for whole journey.
(b) Average speed and Average velocity for whole journey
(c) Average acceleration of cyclist for whole journey.
(Take x and y axis along East and North respectively).

12. A point traversed half the distance with a velocity v_0. The remaining part of the distance was covered with velocity v_1 for half the time, and with velocity v_2 for the other half of the time. Find the mean velocity of the point averaged over the mole time of motion.

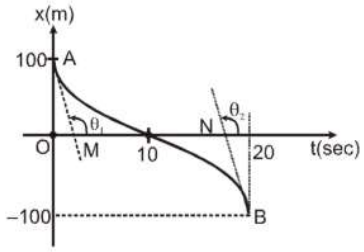
13. A car starts from P and follows the path as shown in figure. Finally car stops at R. Find the distance travelled and displacement of the car if $a = 7m, b = 8m$ and $r = \frac{11}{\pi}m$? (Take $\pi = \frac{22}{7}$)



14. A man moves to go 50 m due south, 40 m due west and 20 m due north to reach a field.
(a) What distance does be have to walk to reach the field?
(b) What is his displacement from his house to the field?

15. A hall has the dimensions 10 m × 10 m × 10 m. A fly starting at one corner ends up at a farthest corner. The magnitude of its displacement is:
(a) $5\sqrt{3}m$ (b) $10\sqrt{3}m$
(c) $20\sqrt{3}m$ (d) $30\sqrt{3}m$

16. **Comprehension #1**
Read the following write up and answer the questions based on that.
The graph below gives the coordinate of a particle travelling along the X-axis as a function of time. AM is the tangent to the curve at the starting moment and BN is tangent at the end moment $(\theta_1 = \theta_2 = 120°)$.



The average velocity during the first 20 seconds is
(a) -10 m/s (b) 10 m/s
(c) Zero (d) 20 m/s

17. A particle is moving in x–y–plane at 2 m/s along x–axis. 2 seconds later, its velocity is 4 m/s in a direction making 60° with positive x–axis. Its average acceleration for this period of motion is:-
(a) $\sqrt{5}m/s^2$, along y-axis
(b) $\sqrt{3}m/s^2$, along y-axis
(c) $\sqrt{5}m/s^2$, along at $60°$ with positive x-axis
(d) $3m/s^2$, at $60°$ with positive x-axis

18. The second's hand of a watch has length 6 cm. Speed of end point and magnitude of difference of velocities at two perpendicular positions will be
(a) 2p & 0 mm/s
(b) $2\sqrt{2}\pi \,\&\, 4.44\,mm/s$
(c) $2\sqrt{2}\pi \,\&\, 2p\,mm/s$
(d) $2p \,\&\, 2\sqrt{2}\pi\,mm/s$

19. An object may have :-
(a) Varying speed without having varying acceleration
(b) Varying velocity without having varying speed
(c) Non–zero acceleration without having varying velocity
(d) Non–zero acceleration without having varying speed

20. Which of the following statements are true for a moving body ?
(a) If its speed changes, its velocity must change and it must have some acceleration
(b) If its velocity changes, its speeds must change and it must have some acceleration
(c) If its velocity changes, its speed may or may not change, and it must have some acceleration
(d) If its speed changes but direction of motion does not change, its velocity may remain constant

21. When a particle is in motion, its acceleration may be in any direction.

22. A particle is rotating in a circle of radius 1m with constant speed 4 m/s. In time 1 s, match the following (in SI units)

Column I	**Column II**
A. Displacement	p. 8 sin 2
B. Distance	q. 4
C. Average velocity	r. 2 sin 2
D. Average acceleration	s. 4 sin 2

ANSWERS

1. (a) 110 m

(b) 50 m, $\tan^{-1}$ 3/4 north to east

2. 60 m, 20 m in the negative direction

3. (a) 520 km/h

(b) 40 km/h

(c) 520 km/h Patna to Ranchi

(d) 32.5 km/h Patna to Ranchi

4. 32 km/h

(b) zero

5. 2.5 m/s^2

6. 48/7 km/hr.

7. 16 ft/min, zero.

8. $2\sqrt{2}$ m/s due north west.

9. (a) $5\sqrt{2}$ due north east (b) 20 + 5 $\sqrt{2}$ m

10. (a) 1100 m

(b) 500 m

(c) $\frac{11}{12} m/s$

(d) $\frac{5}{12} m/s$, 37° due North of East.

11. (a) $\left(25+5\sqrt{2}\right)m$

(b) $5m/s, \frac{-5\hat{j}}{\left(5+\sqrt{2}\right)} m/s$

(c) $\frac{-10\hat{i}+5\hat{j}}{5+\sqrt{2}} m/s^2$

12. $\frac{2v_0\left(v_1+vl2\right)}{2v_0+v_1+v_2}$

13. (Distance travelled by the car = 48 m, Displacement of the car = 36 m)

14. (a) 110 m

(b) 50 m, $\tan^{-1}\frac{4}{3}$ west of south

15. (b)

16. (a)

17. (b)

18. (d)

19. (a, b, d)

20. (a, c)

21. T

22. Straight line